Financial News Sentimental Analysis with Quaternion Transformer

Wenda Zhai (wz2914@nyu.edu)

Summary: As financial sentimental text data plays an increasingly important role in the real market data, this report focuses on the financial sentimental analysis with quaternion transformer, a transformer variant which has high performance with less parameters. By analyzing the relationship between financial news of NVIDIA stock and stock return, we further predict the labeled return by applying financial sentimental analysis in different time period, and we successfully achieved a high accuracy and make predictions for the market.

Key Words: Sentimental Analysis; Financial News; Large Language Model; Quaternion Transformer;

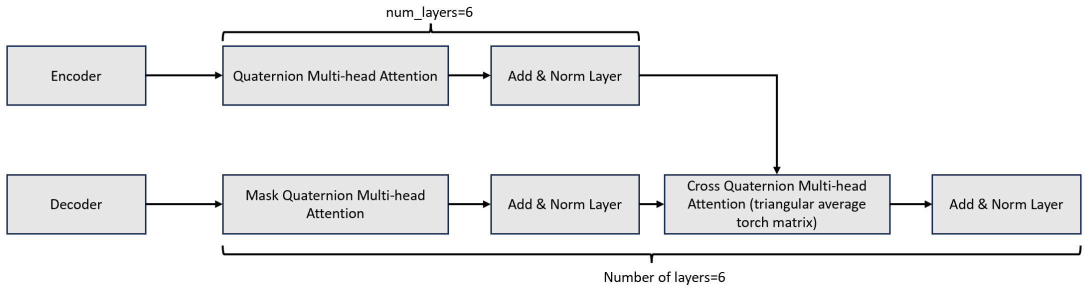
Main Contribution: Training and fun-tuning quaternion transformer model based on hugging face data; Collect and cleanse the financial sentimental news data of NVIDIA; Fine-tune the pre-trained quaternion transformer model for market prediction of NVIDIA stock return; Finish an encoder-decoder quaternion transformer;

Background

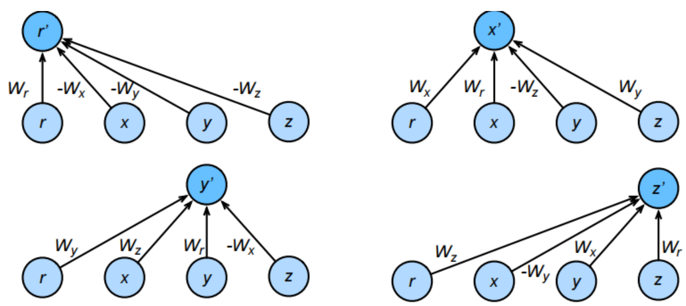
Sentimental News are playing an increasingly important role in the financial worlds. By leveraging complicated large language model build on attention-based algorithm, we can analyze the sentimental text in from website, social media, and financial news and even make predictions. This report will focus on the application of transformer-based models on classifying financial text sentimental data and make market predictions.

Transformer and Quaternion Neural Network

Transformer is an encoder-decoder based model which utilizes the “attention” mechanism. The quaternion transformer, built on the quaternion algebra, replace the Feed Forward Neural Layers (BP layers) are replaced with Quaternion Feed Forward Neural Layers instead of traditional real-valued layers, which can reduce the number of parameters while maintaining performance.



The quaternion transformer model processes real-valued input sequences by grouping them into sets of four, treating them as the four components of a quaternion. After processing, the model interprets the four quaternion components as a standard sequence of real values, making them compatible with traditional structures.



Literature Review of Quaternion Transformer: Quaternion Transformer in Language Translation

A paper (Tay et al., 2019) introduced Quaternion transformer and applied the encoder-decoder quaternion transformer on multiple translation task and achieved the following results:

	BLEU			
Model	IWSLT'15 *En-Vi*	WMT'16 *En-Ro*	WMT'18 *En-Et*	# Params
Transformer Base	28.4	**22.8**	14.1	44M
Quaternion Transformer (*full*)	28.0	18.5	13.1	11M (-75%)
Quaternion Transformer (*partial*)	**30.9**	22.7	**14.2**	29M (-32%)

Here BLEU score (Bilingual Evaluation Understudy) measures the score of language translation. It is the exponential weighted log accuracy scaled by Brevity Penalty (BP):

$$BLEU = BP \exp(\sum_{1}^{N} w_n log p_n)$$

Where w_n is the weight of-n-gram, which is usually assigned equal for each gram, i.e., 0.2 for 5-grams. And the p_n is the accuracy, which is the clipped counting n-gram in the corpus over all count:

$$count_{clip} = \min{(Count, MaxRefCount)}$$

Where the clipped count is the minimum between the actual count and the maximum reference count. And p_n has the following definition:

$$p_n = \frac{\sum_{C\in\{Candidates\}} \sum_{n-gram\in C} Count_{clip}(n-gram)}{\sum_{C'\in\{Candidates\}} \sum_{n-gram'\in C} Count_{clip}(n-gram')}$$

And brevity penalty is the short for length, where r is the reference sentence length, c is the candidate sentence length.

$$BP = \begin{cases} 1, if\ c > r \\ e^{(1-r/c)}, if\ c \leq r \end{cases}$$

N-gram is N-consecutive words. 1-gram measures the precision of word, n-gram measure the fluency of sentence when n>=1. For example, if a sentence is [‘I’, ‘enjoy’,’machine’,’learning’], and the reference dictionary (corpus) is [[‘I’, ‘love’,’machine’,’learning’],[‘I’,’like’,’machine’,’learning’]]. Then if we only consider 1-gram and 2-gram, BP measures the closest length c, so Brevity Penalty is 1. And 3/4 1-gram appears in candidates, 1/3 2-gram appears in candidates. So the BLEU score is:

$$BLEU = 1 * exp\left(0.5 * \left(\log\frac{3}{4} + log\frac{1}{3}\right)\right) \approx 0.5$$

BLEU ranges from 0 to 1. Also, if the word has many duplicates, then the count number will increase, and the accuracy will go down.

In the paper’s result, the quaternion transformer has reduced the parameter. Also, partial quaternion transformer achieved a balance between model size and accuracy.

Noticeably, here the quaternion transformer is encoder-decoder structure, as this is translation task. The encoder-decoder transformer is also finished.

Return Labelling in Prompt Engineering

Yu et al.(2023) used LLMs including Llama, GPT, and traditional ARMA-GARCH and combine historical price data, metadata (information about data) and news data to make summary of news and predict the weekly return of stocks.

Here the historical bin method means, the most frequent bin from historical weeks before the target week to forecast. And the task is to give the news or price as tokens input, concatenated with the stock performance, classified from D5 to U5.

The author assigned the U_i (i=1,2,3,4,5) means price rising between (i − 1)% and i%. Then the author used prompt engineering to do this task. By feeding the encoder-decoder model, different models have the following result.

	Weekly			Monthly (Every 4 Weeks)		
	Binary Precision	**Bin Precision**	**MSE**	**Binary Precision**	**Bin Precision**	**MSE**
Most-Frequent Historical Bin	50.7%	16.4%	43.5	51.4%	17.2%	155.1
ARMA-GARCH	52.4%	11.1%	22.1	50.5%	6.2%	90.1
Gradient Boosting Tree Model	60.8%	26.4%	24.3	56.4%	17.7%	85.6
GPT-4 Zero-Shot	64.5%	31.2%	20.5	64.8%	26.0%	60.1
GPT-4 Few-Shot	65.8%	32.7%	20.6	65.3%	26.5%	58.2
GPT-4 Few-Shot w/ COT	**66.5%**	**35.2%**	**18.7**	**69.5%**	**28.6%**	**50.4**
Open LLaMA (13B) Fine-Tuned	62.2%	26.5%	23.3	60.1%	22.6%	63.3
Open LLaMA (13B) Fine-Tuned w/ COT	64.7%	30.7%	21.0	62.2%	24.4%	63.5

Prompt [template, those inside {} are placeholders]:
Instruction: Forecast next week stock return (price change) for symbol, given the company profile, historical weekly news summary, keywords, and stock returns, and optionally the examples from other stocks of a similar company. The trend is represented by bins "D5+", "D5", "D4", "D3", "D2", "D1", "U1", "U2", "U3", "U4", "U5", "U5+", where "D5+" means price dropping more than 5%, D5 means price dropping between 4% and 5%, "D4" means price dropping between 3% and 4%, "U5+" means price rising more than 5%, "U5" means price rising between 4% and 5%, "D4" means price rising between 3% and 4%, etc.
Company Profile: {company_profile} [company description, general positive/negative factors as shown in Appendix Figure 1.]
Recent News: News are ordered from oldest news to latest news.
====8 Weeks ago====
{meta_news_summary_company} [meta news summary & keywords of the week for the company]
{meta_news_summary_macro} [meta news summary & keywords for the macro economy and finance status of the week] ...
====7 weeks ago====
...
====Last week====
...
Forecasting Examples: {few_shot_learning_examples_from_similar_stocks} [We use forecasting examples from top-3 similar stocks as the few-short learning examples. The example is structured similar to this prompt, with company profile and recent news, but without the instruction texts.]

Now predict what could be the next week's Summary, Keywords, and forecast the Stock Return. The predicted Summary/Keywords should explain the stock return forecasting. You should predict what could happen next week. Do not just summarize the history. The next week stock return need not be the same as the previous week. Use format Summary: ..., Keywords: ..., Stock Return: ...

GPT-4 Response [Forecasting week 05/01/2023 to 05/07/2023]:
Summary: Apple's stock is expected to see fluctuations in the week to come against a backdrop of mixed news. The company's announced expansion into India, coupled with the introduction of high-yield savings accounts, initially provided a boost to the stock. However, growing concerns surrounding reduced PC shipments and apprehensions about the company's upcoming earnings injected a degree of uncertainty. Despite these challenges, Apple's unwavering focus on cost control and the anticipation surrounding innovative offerings, such as the forthcoming mixed-reality headset, shored up investor confidence. Consequently, the stock maintained a general upward trend throughout the week, albeit punctuated by a few dips. The broader economic climate on Wall Street also had a role to play, with a recovering sentiment in Q1 towards high-tech companies like Microsoft and Alphabet. These companies released pivotal financial reports and ground-breaking products during the period, influencing the tech sector's performance, potentially impacting Apple's stock trajectory.

Keywords: AAPL, mixed-reality headset, India expansion, high-yield savings account, cost control strategies, Bernstein, Wall Street, economic recovery, tech sector performance

Stock Return: U1

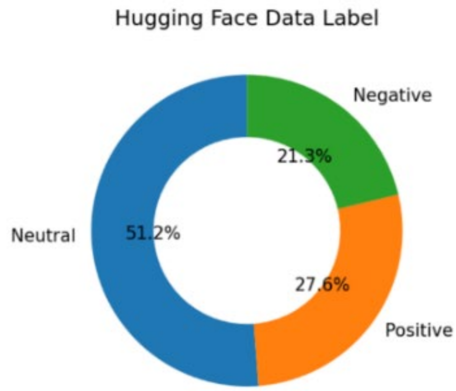
Ground Truth [Summary from actual news from 05/01/2023 to 05/07/2023 and the actual return]:
Summary: Apple's stock saw significant gains after outperforming Wall Street's fiscal second quarter targets, driven by robust iPhone sales and strong services, despite posting a decrease in revenue for the second consecutive quarter and facing potential competition in the VR headset market from Reliance Jio's JioDive. The company also noted an improvement in supply and strength in emerging markets, further boosting investor confidence.

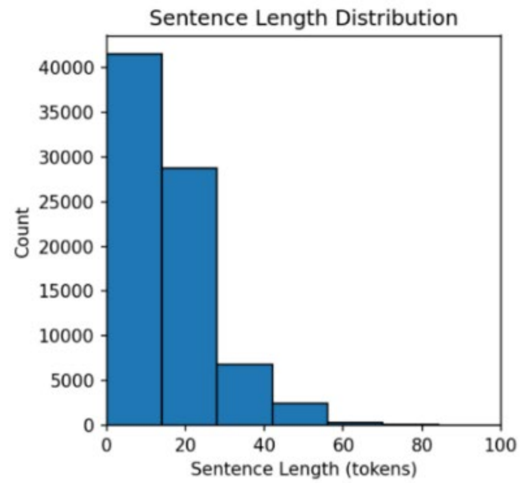
Keywords: Apple, stock, Wall Street, fiscal second quarter, iPhone, services sales, quarterly earnings, revenue, declining revenue, emerging markets, improved supply, Virtual Reality, competition, JioDive, Reliance Jio.

Stock Return: U3

Pre-Trained Model Dataset Description

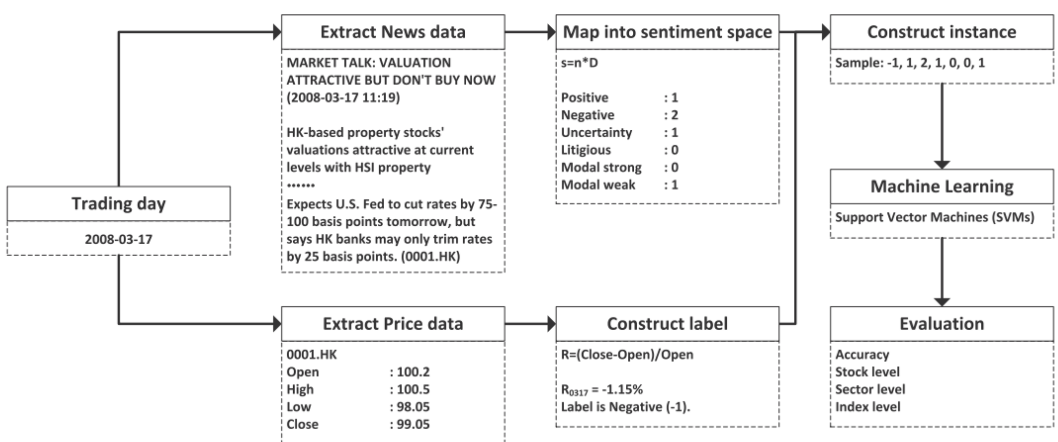
The dataset training the quaternion transformer (which is pretrained model we use later) is a combined dataset from FinPhraseBank, tweet and financial news with 100k rows. Here it shows the distribution of the hugging face. It can be noticed that most of the news pieces are neutral(51.2%), and positive and negative news accounts for 27.6% and 21.3% respectively. And the length of most sentences is less than 60 tokens.





Financial News Analysis

A research was carried on financial news sentimental analysis (Li et al., 2013), the author analyzed the sentiment analysis of stock return based on financial news. The author mapped each day's news into sentimental spaces and do machine learning to fit the stock return.



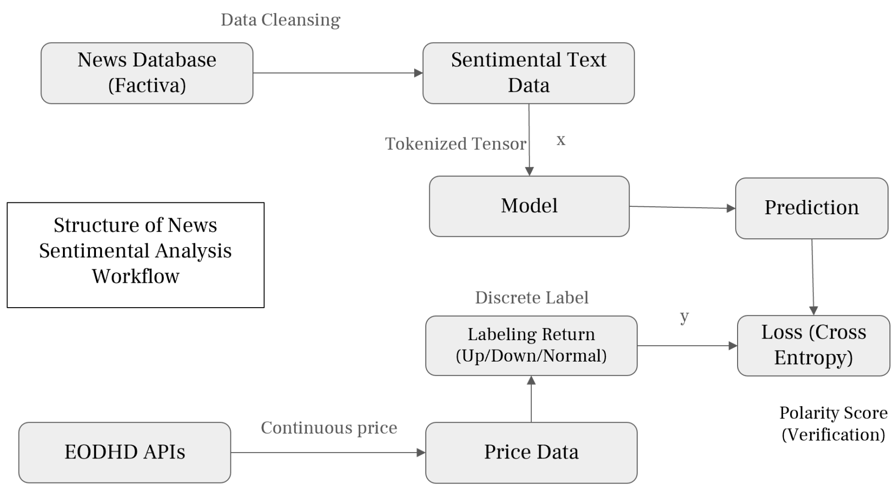
The author extracted all pieces of news in a single day, mapped them into a sentiment space (dictionary), and then align the sentimental with price data label. Then train the machine learning model (SVM), and evaluate the accuracy of the model.

$$L(x) = \begin{cases} positive, if\ R(x) > th \\ neutral, if - th \leq R(x) \leq th \\ negative, if\ R(x) < -th \end{cases}$$

For the threshold, the daily Open-to-Close return is divided into three categories by two thresholds which are symmetric around zero, denoted as th. Here we set the threshold to be 1%. And the prediction label is defined as the formula shown above.

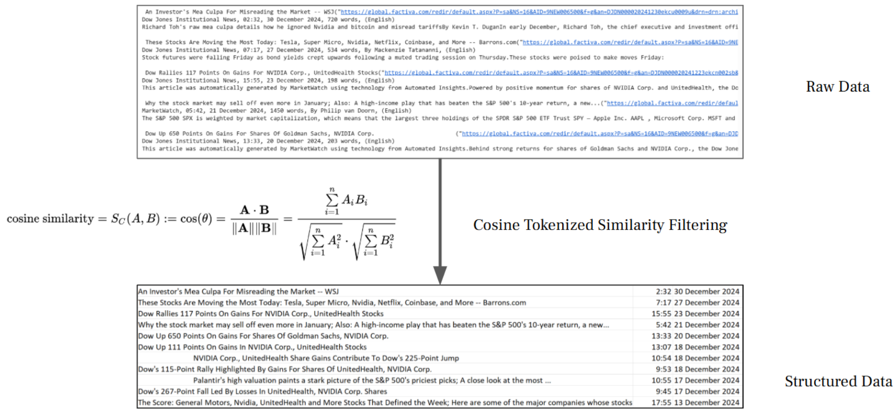
NVIDIA Financial Sentimental News Analysis

Then we move to the financial news analysis of NVIDIA stock, the workflow is shown as below. NVIDIA was chosen because it has the largest amount of news and higher weight in the index as the reference paper shows. Parallel to the paper, we extracted the financial news related to NVIDIA stock from the Factiva database, and used regular expression to do data wrangling. After getting cleansed data, each piece of record should include the title of news, the publish time, and the date of the news. Then we use cross entropy as the loss function of the label of the return. Price data comes from EODHD API (unicorn data services).



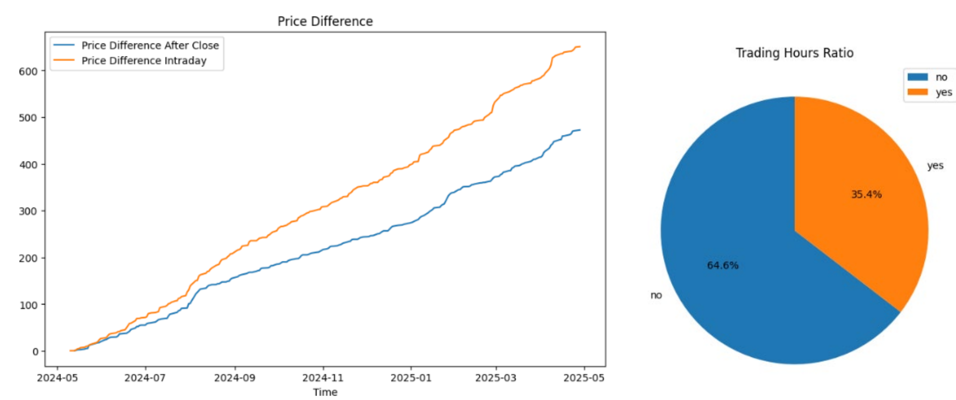
Discussion: Similarity

It is likely that different media sources have similar news. To filter out the duplicates, we use a similarity to filter out all the similar title of news. The similarity is cosine similarity of two vectors based on the count of each word. If two news have very similarity (we set threshold is 0.9), then the only earlier news will be kept. After data cleansing, the size of the data is altogether 3255 effective pieces of news title and 163 test dataset size. The weekly number of news are shown as below.



Discussion: Arrival Time of the News

After getting each piece of news, we delve into the return label of the model. First, we plot the absolute value of price change after close and price change intraday in our dataset from May 2024 to April 2025. The result shows that although the intraday price change is higher, the price change after close (between trading days) also accounts for a large portion of price change. And intraday news accounts for 35.4%, while news after close accounts for 64.6%.



So why does the arrival time of the news matters? Because in non-trading hours, the price will not change and this is the time when the information accumulates. And during the trading time, it is expected the information is reflected in the close price.

So instead of labelling the daily return, we choose to use the intraday return and return after close, and generate $L(x)$ as the label for each news as the formula above.

$$R(x) = \begin{cases} \dfrac{Close_t - Open_t}{Open_t}, \textit{for all news in trading hours} \\ \dfrac{Open_t - Close_{t-1}}{Close_{t-1}}, \textit{for all news after close} \end{cases}$$

Discussion: Multiple News and Polarity Index

The the labelling will lead to a question: what if there are multiple news in the same day with difference sentimental direction? The authors introduced polarity index: that is, number of news classified as positive minus the negative over the sum, to compute a polarity score to compute the market sentiment index and address having multiple news articles per.

$$Polarity\ Index = \frac{positive - negative}{positive + negative}$$

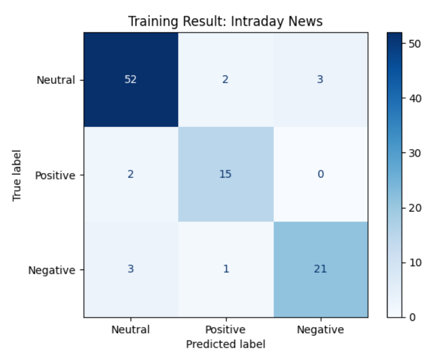
Polarity index ranges from -1 to 1. If the score is 1, it means all the pieces of news are positive. And if it is equal to -1, all pieces of news are negative. Same as the information coefficient (Chuan et al., 2019), the cumulative polarity index can reflect the price trend. So, we can compute the daily polarity index based on our model prediction for each piece of sample, and analyze the cumulative sum.

Result: Model Prediction: Intraday vs After Close

We fine-tune the model based on the quaternion transformer trained so far on the hugging face dataset, and see if we can make predict the label of the stock return. We examined the price change intraday vs after close.

Classification Report:

	precision	recall	f1-score	support
neutral	0.92	0.81	0.86	57
positive	0.78	0.82	0.80	17
negative	0.71	0.88	0.79	25
accuracy			0.83	99
macro avg	0.80	0.84	0.82	99
weighted avg	0.84	0.83	0.83	99



Intraday News Model Accuracy

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Classification Report:
              precision    recall  f1-score   support

     neutral       0.96      0.91      0.94       114
    positive       0.77      0.83      0.80        41
    negative       0.66      0.73      0.69        26

    accuracy                           0.87       181
   macro avg       0.80      0.82      0.81       181
weighted avg       0.88      0.87      0.87       181
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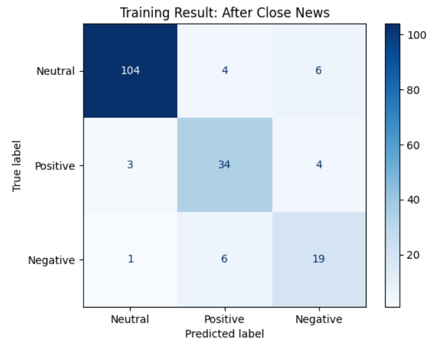
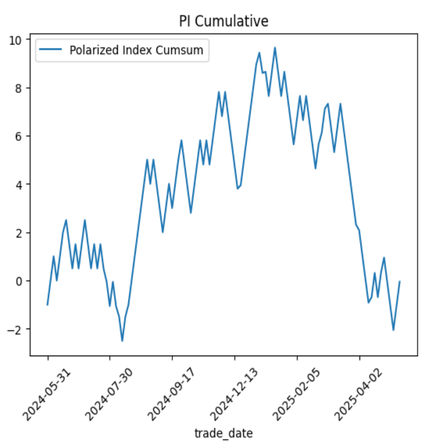


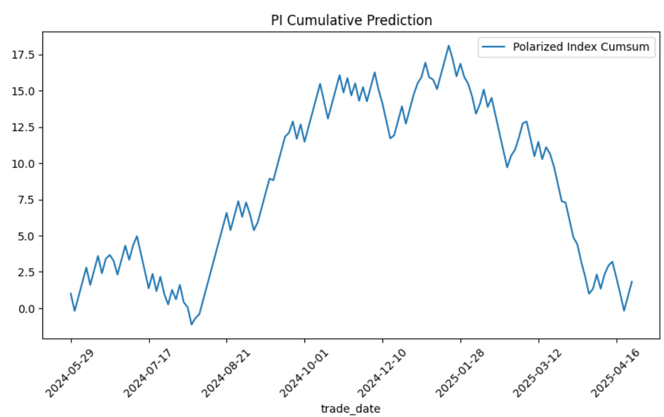
News After Close Model Accuracy

The model has higher test accuracy on the news after close. This probably is owing to the fact that news after close has higher proportion in the training dataset. But overall, the quaternion transformer all achieved a good performance.

Discussion: Cumulative Polarity Index (PI)

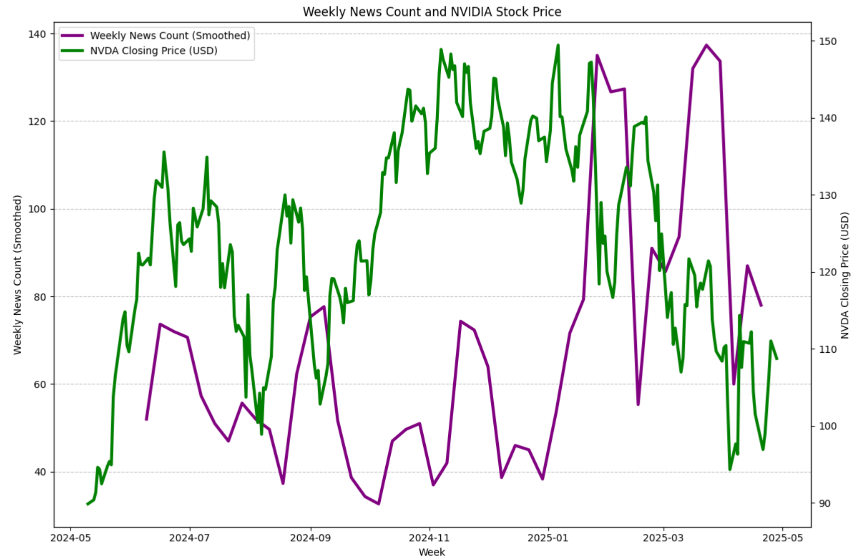
Then we analyze the relationship between news. Because the PI is an imbalance indicator, the original PI, same as the actual trend of the price, is plotted on the left, and the model prediction PI, which reflects the divergence of the news, has a similar trend with real PI. And this reflects exact trend of the price, as the figure of the price showed.





Future Works

Future works can be done in serval ways. First, Asymmetric Feature. This picture shows the closing price and the amount of news in the dataset time range. It can be notice that when the price went up, the number of news didn’t increase a lot. But during 2025 spring drawdown, the number of news increased a lot. Further analysis can focus on the asymmetric property of the news.



In other cases, the change of frequency of the news as well as price data, i.e., if we use higher frequency of the price change (like 1 min return rate for intraday news), the model may have different performance result. In other area, future works can focus on different assets, like how does the model perform on foreign exchange, given that the information of the foreign exchange are highly public. Or how does the model works on the treasury bond's news. In all meanings, this framework of news-price analysis should work on any assets.

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